

SECTION 1: DOUBLE PARALLEL FILLET WELD

System Parameters: Plate Width $b = 100\text{ mm}$, Thickness $s = 10\text{ mm}$, Static Load $P = 80\text{ kN} = 80 \times 10^3\text{ N}$, Permissible Shear Stress $\tau = 55\text{ MPa} = 55\text{N/mm}^2$.

Design Protocol: Calculate effective weld length l for double parallel fillet weld under static load and total weld run length L adding 12.5 mm start/stop allowance.

1. Effective Length of Weld (l) Double parallel fillet weld load capacity	$P = 1.414 \cdot s \cdot l \cdot \tau$ $80 \times 10^3 = 1.414 \times 10 \times l \times 55$ $l = \frac{80 \times 10^3}{778}$ $= 102.8\text{ mm}$ $\Rightarrow l = \mathbf{103\text{ mm}}$
2. Total Length of Weld Run (L) Adding 12.5 mm allowance for starting and stopping	$L = l + 12.5\text{ mm}$ $= 103 + 12.5$ $= \mathbf{115.5\text{ mm}}$
3. Design Output Summary Sized Double Parallel Fillet Weld	Total Length of Weld Run $L = 115.5\text{ mm}$

SECTION 2: CIRCULAR FILLET WELD UNDER TORSION

System Parameters: Shaft Diameter $d = 50\text{ mm}$, Weld Size $s = 10\text{ mm}$, Allowable Shear Stress $\tau_{\text{max}} = 80\text{ MPa} = 80\text{ N/mm}^2$.

Design Protocol: Determine maximum torque capacity T of circular fillet weld joining solid shaft to flat plate.

1. Maximum Torque Capacity (T)
Torsional shear stress in circular fillet weld

$$\begin{aligned}\tau_{\text{max}} &= \frac{2.83T}{\pi \cdot s \cdot d^2} \\ 80 &= \frac{2.83T}{\pi \times 10 \times (50)^2} \\ 80 &= \frac{2.83T}{78550} \\ T &= \frac{80 \times 78550}{2.83} \\ &= 2.22 \times 10^6\text{ N} \cdot \text{mm} \\ &= 2.22\text{ kN} \cdot \text{m}\end{aligned}$$

2. Design Output Summary
Circular Fillet Weld Torque Capacity

Maximum Torque Capacity $T = 2.22\text{ kN.m}$

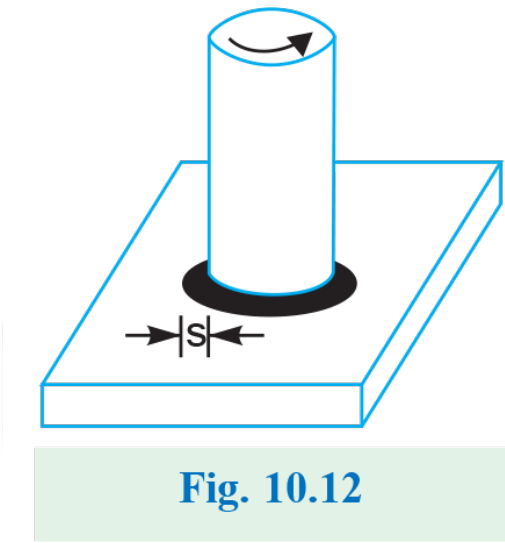


Figure 10.12: Solid shaft welded to plate by circular fillet weld under torsion

SECTION 3: LONG FILLET WELD UNDER TORSION

System Parameters: Plate Length $l = 1\text{ m} = 1000\text{ mm}$, Thickness = 60 mm, Weld Size $s = 15\text{ mm}$, Permissible Shear Stress $\tau_{\text{max}} = 80\text{ MPa} = 80\text{ N/mm}^2$.

Design Protocol: Calculate maximum torque capacity T for long fillet weld connecting two plates at right angles.

1. Maximum Torque Capacity (T)
Torsional shear stress in long fillet weld

$$\begin{aligned}\tau_{\text{max}} &= \frac{4.242T}{s \cdot l^2} \\ 80 &= \frac{4.242T}{15 \times (1000)^2} \\ 80 &= \frac{0.283T}{10^6} \\ T &= \frac{80 \times 10^6}{0.283} \\ &= 283 \times 10^6\text{ N} \cdot \text{mm} \\ &= 283\text{ kN} \cdot \text{m}\end{aligned}$$

2. Design Output Summary
Long Fillet Weld Torque Capacity

Maximum Torque Capacity $T = 283\text{ kN.m}$

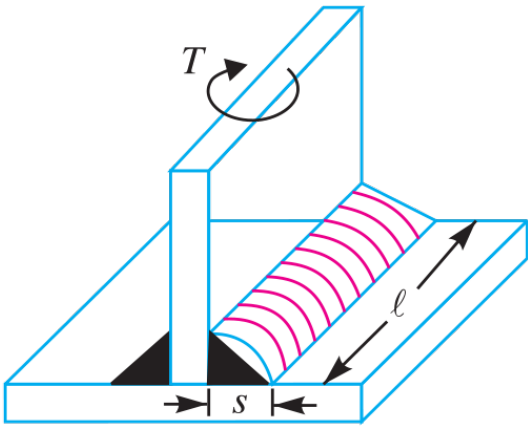


Fig. 10.13

Figure 10.13: Long fillet weld subjected to torsion

SECTION 4: PARALLEL FILLET WELD — STATIC & FATIGUE LOADING

System Parameters: Plate Width $b = 100$ mm, Weld Size $s = 12.5$ mm, Load $P = 50$ kN $= 50 \times 10^3$ N, Allowable Shear Stress $\tau = 56$ MPa, Stress Concentration Factor $K_f = 2.7$.

Design Protocol: Calculate required weld run length under static loading and fatigue loading.

1. Length of Weld for Static Loading (l) Load capacity for double parallel fillet weld	$P = 1.414 \cdot s \cdot l \cdot \tau$$50 \times 10^3 = 1.414 \times 12.5 \times l \times 56$$l = \frac{50 \times 10^3}{990}$$= 50.5 \text{ mm}$
2. Total Static Weld Length (L) Adding 12.5 mm start/stop allowance	$L = l + 12.5 \text{ mm}$$= 50.5 + 12.5$$= \mathbf{63 \text{ mm}}$
3. Permissible Shear Stress under Fatigue (τ_{fatigue}) Fatigue stress concentration factor $K_f = 2.7$	$\tau_{\text{fatigue}} = \frac{\tau}{K_f}$$= \frac{56}{2.7}$$= 20.74 \text{ N/mm}^2$
4. Total Fatigue Weld Length (L_{fatigue}) Effective length and total run length under fatigue loading	$P = 1.414 \cdot s \cdot l_{\text{fatigue}} \cdot \tau_{\text{fatigue}}$$50 \times 10^3 = 1.414 \times 12.5 \times l_{\text{fatigue}} \times 20.74$$l_{\text{fatigue}} = \frac{50 \times 10^3}{367}$$= 136.2 \text{ mm}$$L_{\text{fatigue}} = 136.2 + 12.5$$= \mathbf{148.7 \text{ mm}}$
5. Design Output Summary Parallel Fillet Weld Lengths	Static Length $L = 63$ mm, Fatigue Length $L_{\text{fatigue}} = \mathbf{148.7 \text{ mm}}$

SECTION 5: COMBINATION WELD — STATIC & FATIGUE LOADING

System Parameters: Width $b = 75$ mm, Thickness $s = 12.5$ mm, Tensile Stress $\sigma_t = 70$ MPa, Shear Stress $\tau = 56$ MPa, $K_{fl} = 1.5$ (transverse), $K_{p2} = 2.7$ (parallel).

Design Protocol: Determine length of each parallel fillet weld for both static and fatigue loading.

1. Total Plate Load Capacity (P) Maximum tensile load plate can carry	$P = b \cdot s \cdot \sigma_t$$= 75 \times 12.5 \times 70$$= 65625 \text{ N}$
2. Transverse Weld Effective Length (l_1) Width minus start/stop allowance	$l_1 = b - 12.5 \text{ mm}$$= 75 - 12.5$$= 62.5 \text{ mm}$
3. Parallel Fillet Length for Static Loading (l_2) Load sharing between transverse and parallel welds	$P = P_1 + P_2$$P_1 = 0.707s \cdot l_1 \cdot \sigma_t$$= 0.707 \times 12.5 \times 62.5 \times 70 = 38664 \text{ N}$$P_2 = 1.414s \cdot l_2 \cdot \tau$$= 1.414 \times 12.5 \times l_2 \times 56 = 990l_2$$65625 = 38664 + 990l_2$$l_2 = \frac{65625 - 38664}{990}$$= 27.2 \text{ mm}$$\Rightarrow l_2 + 12.5 = 40 \text{ mm}$
4. Parallel Fillet Length for Fatigue Loading (l_2, fatigue) Fatigue permissible stresses and load balancing	$\sigma_{t, \text{ fatigue}} = \frac{70}{1.5} = 46.7\text{N/mm}^2$$\tau_{\text{fatigue}} = \frac{56}{2.7} = 20.74\text{N/mm}^2$$P_1 = 0.707(12.5)(62.5)(46.7) = 25795 \text{ N}$$P_2 = 1.414(12.5)l_2(20.74) = 366l_2$$65625 = 25795 + 366l_2$$l_2 = \frac{65625 - 25795}{366}$$= 108.8 \text{ mm}$$\Rightarrow l_2 + 12.5 = 121.3 \text{ mm}$
5. Design Output Summary Combination Weld Lengths	Static Parallel Weld l2 = 40 mm, Fatigue Parallel Weld l2 = 121.3 mm

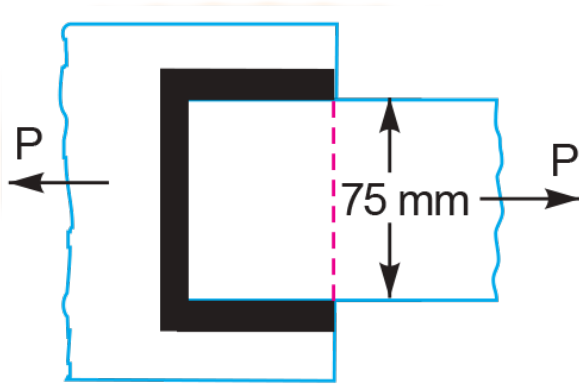


Fig. 10.15

Figure 10.15: Combination of single transverse and double parallel fillet weld

SECTION 6: WELD RUN LENGTH UNDER VARIABLE LOADING

System Parameters: Plate Width $b = 120$ mm, Thickness $s = 15$ mm, Static Tensile Stress $\sigma_t = 70$ MPa, Static Shear Stress $\tau = 56$ MPa, $K_{f1} = 1.5$, $K_{f2} = 2.7$.

Design Protocol: Determine weld run lengths for single transverse weld and double parallel fillet welds.

1. Single Transverse Weld Length (l_1) Width minus start/stop allowance	$\begin{aligned} l_1 &= b - 12.5 \text{ mm} \\ &= 120 - 12.5 \\ &= 107.5 \text{ mm} \end{aligned}$
2. Double Parallel Fillet Weld Length (l_2) Plate load capacity and fatigue load balance	$\begin{aligned} P &= b \cdot s \cdot \sigma_t \\ &= 120 \times 15 \times 70 \\ &= 126 \times 10^3 \text{ N} \\ P_1 &= 0.707(15)(107.5)(46.7) = 53240 \text{ N} \\ P_2 &= 1.414(15)l_2(20.74) = 440l_2 \\ 126 \times 10^3 &= 53240 + 440l_2 \\ l_2 &= \frac{126000 - 53240}{440} \\ &= 165.4 \text{ mm} \\ \Rightarrow l_2 + 12.5 &= 178 \text{ mm} \end{aligned}$
3. Design Output Summary Dynamic Load Weld Run Lengths	Transverse Weld Length l1 = 107.5 mm, Double Parallel Fillet Length l2 = 178 mm

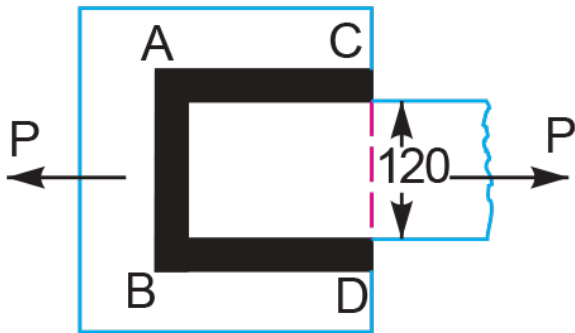


Fig. 10.16

Figure 10.16: Long fillet weld subjected to dynamic variable loading

SECTION 7: FILLET WELD 'T' SECTION SHEAR PLANE ANALYSIS

System Parameters: Leg size s , Weld length l .

Design Protocol: Locate maximum shear plane angle θ for parallel and transverse loading, and find ratio of limiting loads.

1. Shear Plane for Parallel Loading (θ)

Maximizing shear stress by differentiation

$$\begin{aligned}\tau &= \frac{P(\cos \theta + \sin \theta)}{2s \cdot l} \\ \frac{d\tau}{d\theta} &= \frac{P}{2s \cdot l}(-\sin \theta + \cos \theta) = 0 \\ \sin \theta &= \cos \theta \\ \Rightarrow \theta &= 45^{\text{deg}} \\ P_{\text{parallel}} &= 1.414 \cdot s \cdot l \cdot \tau_{\text{max}}\end{aligned}$$

2. Shear Plane for Transverse Loading (θ)

Normal and shear force equilibrium resolution

$$\begin{aligned}\tau &= \frac{P \sin \theta (\cos \theta + \sin \theta)}{2s \cdot l} \\ \frac{d\tau}{d\theta} &= 0 \\ \cos 2\theta + \sin 2\theta &= 0 \\ \tan 2\theta &= -1 \\ 2\theta &= 135^{\text{deg}} \\ \Rightarrow \theta &= 67.5^{\text{deg}} \\ P_{\text{transverse}} &= 1.65 \cdot s \cdot l \cdot \tau_{\text{max}}\end{aligned}$$

3. Ratio of Limiting Loads

Ratio of parallel load capacity to transverse load capacity

$$\begin{aligned}\text{Ratio} &= \frac{P_{\text{parallel}}}{P_{\text{transverse}}} \\ &= \frac{1.414s \cdot l \cdot \tau_{\text{max}}}{1.65s \cdot l \cdot \tau_{\text{max}}} \\ &= 0.857\end{aligned}$$

4. Design Output Summary

T-Section Shear Planes

Parallel Plane theta = 45 deg, Transverse Plane theta = 67.5 deg, Load Ratio = 0.857

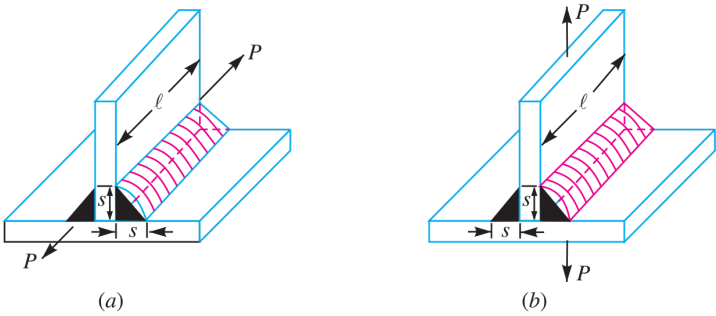


Fig. 10.17

Figure 10.17: Fillet welds fabricating a T-section under parallel and transverse loads

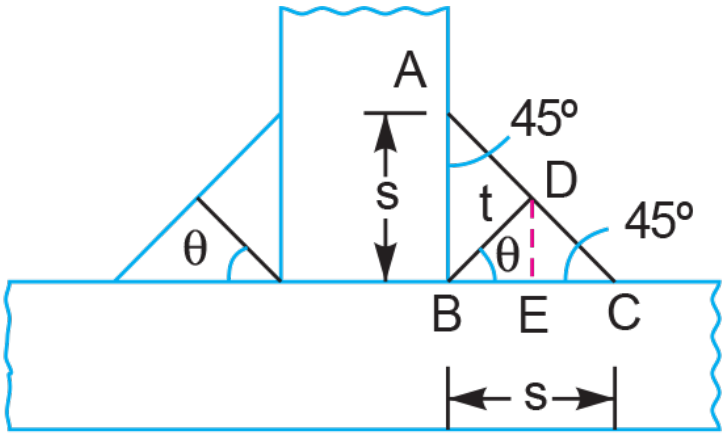


Fig. 10.18

Figure 10.18: Throat plane dimensions for load parallel to weld

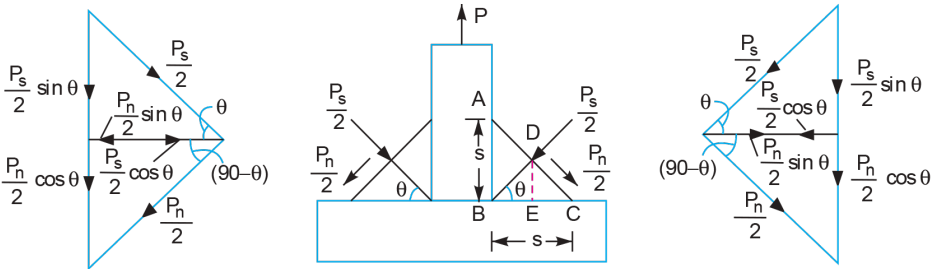


Fig. 10.19

Figure 10.19: Normal and shear force components for transverse load

SECTION 8: AXIALLY LOADED UNSYMMETRICAL SECTION ANGLE WELDING

System Parameters: Angle $200 \times 150 \times 10$ mm ($a + b = 200$ mm), Static Load $P = 200$ kN $= 200 \times 10^3$ N, Allowable Shear Stress $\tau = 75$ MPa, Weld Size $s = 10$ mm.

Design Protocol: Determine top (l_a) and bottom (l_b) weld lengths by balancing load about centroidal axis.

1. Total Required Weld Length (l) Load capacity for single parallel fillet welds	$P = 0.707 \cdot s \cdot l \cdot \tau$ $200 \times 10^3 = 0.707 \times 10 \times l \times 75$ $l = \frac{200 \times 10^3}{530.25}$ $= 377 \text{ mm}$
2. Position of Centroidal Axis (a, b) Distance of gravity axis from bottom edge	$b = \frac{(200 - 10) \times 10 \times 95 + 150 \times 10 \times 5}{190 \times 10 + 150 \times 10}$ $= \frac{188000}{3400}$ $= 55.3 \text{ mm}$ $a = 200 - 55.3 = 144.7 \text{ mm}$
3. Top and Bottom Weld Lengths (l_a, l_b) Moment equilibrium about centroidal line	$l_a = l \cdot \left(\frac{b}{a + b} \right)$ $= 377 \times \left(\frac{55.3}{200} \right)$ $= 104.2 \text{ mm}$ $l_b = 377 - 104.2$ $= 272.8 \text{ mm}$
4. Design Output Summary Unsymmetrical Angle Weld Lengths	Top Weld Length $l_a = 104.2$ mm, Bottom Weld Length $l_b = 272.8$ mm

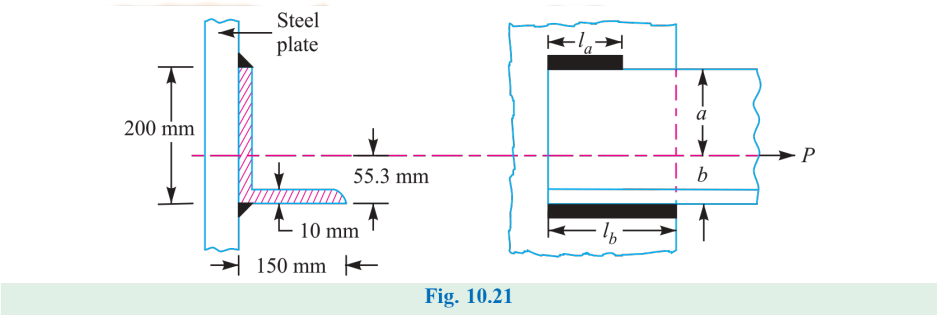


Figure 10.21: Unsymmetrical angle section welded to plate with top and bottom welds

SECTION 9: ECCENTRICALLY LOADED OUT-OF-PLANE WELDED JOINT

System Parameters: Load $P = 2 \text{ kN} = 2000 \text{ N}$, Eccentricity $e = 120 \text{ mm}$, Weld Length $l = 40 \text{ mm}$, Allowable Max Shear Stress $\tau_{\text{max}} = 25 \text{ MPa}$.

Design Protocol: Determine weld size s under combined direct shear and bending.

1. Direct Shear Stress (τ) Throat area of double fillet weld	$\tau = \frac{P}{A} = \frac{2000}{1.414s \cdot l}$$= \frac{2000}{1.414s \times 40}$$= \frac{35.4}{s} \text{N/mm}^2$
2. Bending Stress (σ_b) Bending moment and throat section modulus	$\sigma_b = \frac{M}{Z} = \frac{P \cdot e}{\frac{s \cdot l^2}{4.242}}$$= \frac{2000 \times 120}{\frac{s \times (40)^2}{4.242}}$$= \frac{636.6}{s} \text{N/mm}^2$
3. Maximum Shear Stress & Weld Size (s) Maximum shear stress theory	$\tau_{\text{max}} = \frac{1}{2} \sqrt{\sigma_b^2 + 4\tau^2}$$25 = \frac{1}{2} \sqrt{\left(\frac{636.6}{s}\right)^2 + 4\left(\frac{35.4}{s}\right)^2}$$25 = \frac{320.3}{s}$$s = \frac{320.3}{25}$$= 12.8 \text{ mm}$
4. Design Output Summary Out-of-Plane Joint Weld Size	Required Weld Size $s = 12.8 \text{ mm}$

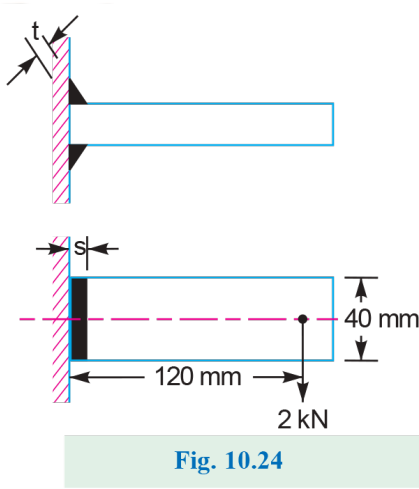


Fig. 10.24

Figure 10.24: Joint subjected to out-of-plane eccentric load

SECTION 10: CIRCULAR FILLET WELD UNDER ECCENTRIC BENDING LOAD

System Parameters: Shaft Diameter $D = 50$ mm, Weld Size $s = 15$ mm, Load $P = 10$ kN = 10000 N, Eccentricity $e = 200$ mm.

Design Protocol: Determine maximum normal and shear stresses in circular fillet weld.

1. Direct Shear Stress (τ) Throat area for circular weld	$\tau = \frac{P}{0.707s \cdot \pi D}$$= \frac{10000}{0.707 \times 15 \times \pi \times 50}$$= 6\text{N/mm}^2 = 6 \text{ MPa}$
2. Bending Stress (σ_b) Bending moment and circular section modulus	$\sigma_b = \frac{M}{Z} = \frac{P \cdot e}{\frac{\pi}{4}(0.707s)D^2}$$= \frac{10000 \times 200}{\frac{\pi}{4}(0.707 \times 15)(50)^2}$$= \frac{2 \times 10^6}{20825}$$= 96\text{N/mm}^2 = 96 \text{ MPa}$
3. Maximum Normal Stress ($\sigma_{t(\text{max})}$) Principal stress theory	$\sigma_{t(\text{max})} = \frac{1}{2}\sigma_b + \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2}$$= 48 + \frac{1}{2}\sqrt{(96)^2 + 4(6)^2}$$= 48 + 48.4$$= \mathbf{96.4 \text{ MPa}}$
4. Maximum Shear Stress (τ_{max}) Maximum shear stress theory	$\tau_{\text{max}} = \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2}$$= \frac{1}{2}\sqrt{(96)^2 + 4(6)^2}$$= \mathbf{48.4 \text{ MPa}}$
5. Design Output Summary Circular Weld Stresses	Max Normal Stress = 96.4 MPa, Max Shear Stress = 48.4 MPa

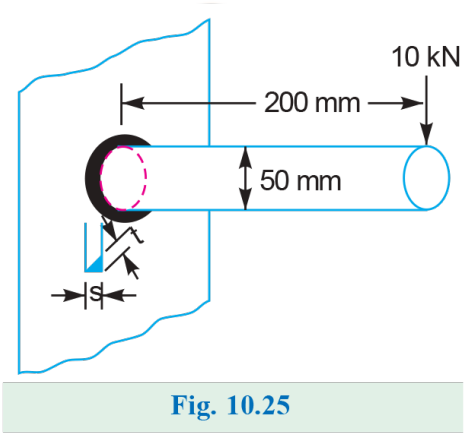


Figure 10.25: Solid shaft welded to plate subjected to eccentric bending load

SECTION 11: RECTANGULAR CROSS-SECTION BAR WELDED TO SUPPORT

System Parameters: Load $P = 25 \text{ kN} = 25 \times 10^3 \text{ N}$, Eccentricity $e = 500 \text{ mm}$, $l = 100 \text{ mm}$, $b = 150 \text{ mm}$, Permissible Shear Stress $\tau_{\text{max}} = 75 \text{ MPa}$.

Design Protocol: Calculate weld size s for rectangular bar welded on 4 sides.

1. Direct Shear Stress (τ) Throat area around all 4 sides	$\tau = \frac{P}{0.707s(2b + 2l)}$$= \frac{25 \times 10^3}{0.707s(2 \times 150 + 2 \times 100)}$$= \frac{70.72}{s} \text{N/mm}^2$
2. Bending Stress (σ_b) Bending moment and throat section modulus	$\sigma_b = \frac{M}{Z} = \frac{P \cdot e}{0.707s\left(\frac{b^2}{3} + b \cdot l\right)}$$= \frac{25 \times 10^3 \times 500}{0.707s\left(\frac{(150)^2}{3} + 150 \times 100\right)}$$= \frac{12.5 \times 10^6}{15907.5s}$$= \frac{785.8}{s} \text{N/mm}^2$
3. Maximum Shear Stress & Weld Size (s) Maximum shear stress theory	$\tau_{\text{max}} = \frac{1}{2} \sqrt{\sigma_b^2 + 4\tau^2}$$75 = \frac{1}{2} \sqrt{\left(\frac{785.8}{s}\right)^2 + 4\left(\frac{70.72}{s}\right)^2}$$75 = \frac{399.2}{s}$$s = \frac{399.2}{75}$$= 5.32 \text{ mm}$
4. Design Output Summary Rectangular Bar Weld Size	Required Weld Size $s = 5.32 \text{ mm}$

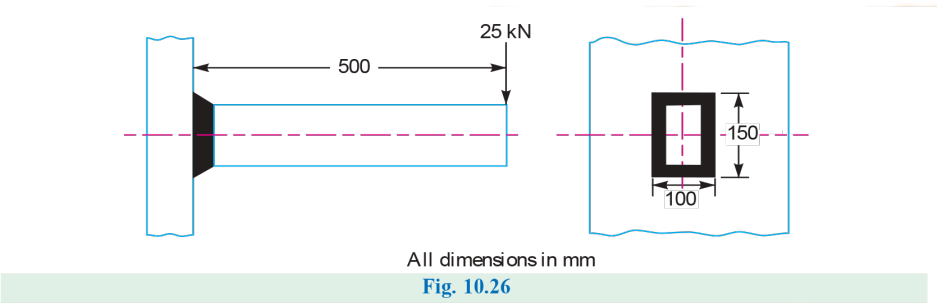


Figure 10.26: Rectangular cross-section bar welded on 4 sides subjected to eccentric load

SECTION 12: ARM WELDED TO HOLLOW SHAFT UNDER BENDING AND TORSION

System Parameters: Force $P = 15 \text{ kN} = 15 \times 10^3 \text{ N}$, Shaft Diameter $d = 80 \text{ mm}$, Torque Arm 240 mm , Bending Arm 175 mm , Permissible Shear Stress $\tau_{\text{max}} = 120 \text{ MPa}$.

Design Protocol: Determine weld size s for combined torsion and bending.

1. Torsional Shear Stress (τ) Torque acting on shaft and torsional shear	$\tau = \frac{2.83T}{\pi \cdot s \cdot d^2}$$T = 15 \times 10^3 \times 240 = 3600 \times 10^3 \text{ N} \cdot \text{mm}$$\tau = \frac{2.83 \times 3600 \times 10^3}{\pi \cdot s \times (80)^2}$$= \frac{506.6}{s} \text{ N/mm}^2$
2. Bending Stress (σ_b) Bending moment and bending stress	$\sigma_b = \frac{5.66M}{\pi \cdot s \cdot d^2}$$M = 15 \times 10^3 \times (200 - 25) = 2625 \times 10^3 \text{ N} \cdot \text{mm}$$\sigma_b = \frac{5.66 \times 2625 \times 10^3}{\pi \cdot s \times (80)^2}$$= \frac{738.8}{s} \text{ N/mm}^2$
3. Maximum Shear Stress & Weld Size (s) Maximum shear stress theory	$\tau_{\text{max}} = \frac{1}{2} \sqrt{\sigma_b^2 + 4\tau^2}$$120 = \frac{1}{2} \sqrt{\left(\frac{738.8}{s}\right)^2 + 4\left(\frac{506.6}{s}\right)^2}$$120 = \frac{627}{s}$$s = \frac{627}{120}$$= 5.2 \text{ mm}$
4. Design Output Summary Arm to Shaft Weld Size	Required Weld Size $s = 5.2 \text{ mm}$

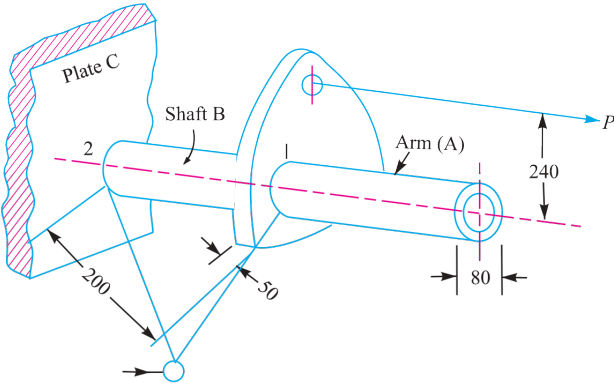


Fig. 10.27. All dimensions in mm.

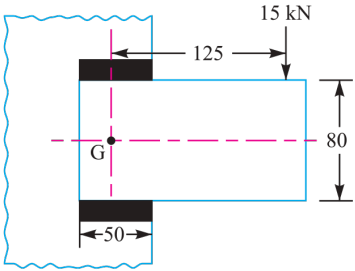
Figure 10.27: Arm welded to hollow shaft subjected to combined bending and torsion

SECTION 13: ECCENTRICALLY LOADED IN-PLANE BRACKET WELD

System Parameters: Load $P = 15 \text{ kN} = 15 \times 10^3 \text{ N}$, $\tau = 80 \text{ MPa}$, $b = 80 \text{ mm}$, $l = 50 \text{ mm}$, Eccentricity $e = 125 \text{ mm}$.

Design Protocol: Determine weld size s under primary direct shear and secondary torsional shear.

1. Primary Direct Shear Stress (τ_1) Throat area and direct shear	$\tau_1 = \frac{P}{2 \cdot 0.707s \cdot l}$$= \frac{15 \times 10^3}{1.414s \times 50}$$= \frac{212}{s} \text{N/mm}^2$
2. Secondary Shear Stress (τ_2) Polar moment of inertia and maximum radius	$J = \frac{0.707s \cdot l(3b^2 + l^2)}{6}$$J = \frac{0.707s \times 50[3(80)^2 + (50)^2]}{6} = 127850s \text{mm}^4$$r_2 = \sqrt{(40)^2 + (25)^2} = 47 \text{ mm}$$\cos \theta = \frac{25}{47} = 0.532$$\tau_2 = \frac{P \cdot e \cdot r_2}{J}$$= \frac{15 \times 10^3 \times 125 \times 47}{127850s} = \frac{689.3}{s} \text{N/mm}^2$
3. Resultant Shear Stress & Weld Size (s) Vector addition of primary and secondary shear	$\tau = \sqrt{\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta}$$80 = \sqrt{\left(\frac{212}{s}\right)^2 + \left(\frac{689.3}{s}\right)^2 + 2\left(\frac{212}{s}\right)\left(\frac{689.3}{s}\right)(0.532)}$$80 = \frac{822}{s}$$s = \frac{822}{80}$$= 10.3 \text{ mm}$
4. Design Output Summary In-Plane Bracket Weld Size	Required Weld Size $s = 10.3 \text{ mm}$



All dimensions in mm.

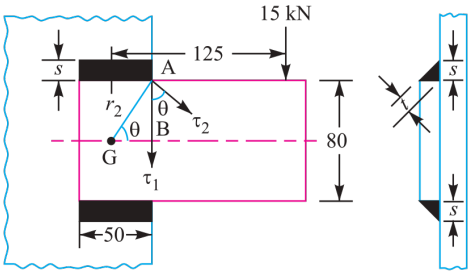


Fig. 10.28

Fig. 10.29

Figure 10.28 & 10.29: Bracket carrying eccentric load and polar centroid vector diagram

SECTION 14: CANTILEVER PLATE CHANNEL WELD UNDER ECCENTRIC LOADING

System Parameters: Load $P = 60 \text{ kN} = 60 \times 10^3 \text{ N}$, $b = 100 \text{ mm}$, $l = 50 \text{ mm}$, Allowable Shear Stress $\tau = 140 \text{ MPa}$.

Design Protocol: Determine weld size s for channel section weld group under eccentric load.

1. Centroid (G) & Polar Moment of Inertia (J) Centroid distance x and polar moment	$x = \frac{l^2}{2l + b}$$= \frac{(50)^2}{2(50) + 100} = 12.5 \text{ mm}$$J = 0.707s \left[\frac{(100 + 100)^3}{12} - \frac{(50)^2(150)^2}{200} \right]$$= 275 \times 10^3 s \text{ mm}^4$
2. Eccentricity (e) & Critical Radius (r_2) Distance to load and critical corner radius	$e = 150 + 50 - 12.5 = 187.5 \text{ mm}$$r_2 = \sqrt{(50)^2 + (37.5)^2} = 62.5 \text{ mm}$$\cos \theta = \frac{37.5}{62.5} = 0.6$
3. Primary & Secondary Shear Stresses (τ_1, τ_2) Direct and torsional shear components	$\tau_1 = \frac{P}{A}, \quad \tau_2 = \frac{P \cdot e \cdot r_2}{J}$$\tau_1 = \frac{60 \times 10^3}{141.4s} = \frac{424}{s} \text{ N/mm}^2$$\tau_2 = \frac{60 \times 10^3 \times 187.5 \times 62.5}{275 \times 10^3 s} = \frac{2557}{s} \text{ N/mm}^2$
4. Resultant Shear Stress & Weld Size (s) Vector combination of shear stresses	$\tau = \sqrt{\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta}$$140 = \sqrt{\left(\frac{424}{s}\right)^2 + \left(\frac{2557}{s}\right)^2 + 2\left(\frac{424}{s}\right)\left(\frac{2557}{s}\right)(0.6)}$$140 = \frac{2832}{s}$$s = \frac{2832}{140}$$= 20.23 \text{ mm}$
5. Design Output Summary Channel Cantilever Weld Size	Required Weld Size $s = 20.23 \text{ mm}$

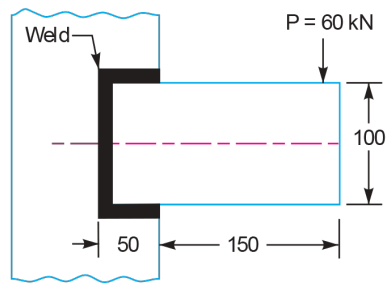


Fig. 10.30

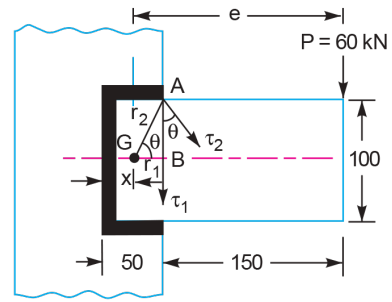


Fig. 10.31

All dimensions in mm.

Figure 10.30 & 10.31: Cantilever plate welded with channel weld pattern and centroid G

SECTION 15: CHANNEL SECTION WELDED TO PLATE UNDER ECCENTRIC IN-PLANE LOADING

System Parameters: Weld Size $s = 6 \text{ mm}$, Load $P = 20 \text{ kN} = 20 \times 10^3 \text{ N}$, $l = 40 \text{ mm}$, $b = 90 \text{ mm}$, Distance to load = 200 mm .

Design Protocol: Determine maximum shear stress $\tau_{\max}$ induced in 6 mm weld.

1. Centroid (G) & Polar Moment of Inertia (J) Location of C.G. and polar moment of throat	$x = \frac{l^2}{2l + b}$ $= \frac{(40)^2}{2(40) + 90} = 9.4 \text{ mm}$ $J = 0.707(6) \left[\frac{(90 + 80)^3}{12} - \frac{(40)^2(130)^2}{170} \right]$ $= 1062.2 \times 10^3 \text{ mm}^4$
2. Eccentricity (e) & Critical Radius (r_2) Load arm and radius to farthest weld point	$e = 200 - 9.4 = 190.6 \text{ mm}$ $r_2 = \sqrt{(45)^2 + (30.6)^2} = 54.4 \text{ mm}$ $\cos \theta = \frac{30.6}{54.4} = 0.5625$
3. Primary & Secondary Shear Stresses (τ_1, τ_2) Direct and torsional shear components	$\tau_1 = \frac{P}{A}, \quad \tau_2 = \frac{P \cdot e \cdot r_2}{J}$ $\tau_1 = \frac{20 \times 10^3}{721.14} = 27.7 \text{ N/mm}^2$ $\tau_2 = \frac{20 \times 10^3 \times 190.6 \times 54.4}{1062.2 \times 10^3} = 195.2 \text{ N/mm}^2$
4. Maximum Resultant Shear Stress ($\tau_{\max}$) Resultant shear stress in 6 mm weld	$\tau_{\max} = \sqrt{\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta}$ $= \sqrt{(27.7)^2 + (195.2)^2 + 2(27.7)(195.2)(0.5625)}$ $= \sqrt{44903.6}$ $= 212 \text{ MPa}$
5. Design Output Summary Channel Weld Maximum Stress	Maximum Shear Stress tau_max = 212 MPa



Figure 10.32: Channel section welded to plate and loaded at 200 mm distance



Figure 10.33: Location of centroid G and critical radius r2

SECTION 16: SHIELDED ARC WELDED BRACKET UNDER DEAD WEIGHT

System Parameters: Dead Weight $P = 15\text{ kN} = 15 \times 10^3\text{ N}$, Permissible Strength $\tau = 150\text{ MPa}$, Weld Length $l = 25\text{ mm}$.

Design Protocol: Calculate top (s_1) and bottom (s_2) fillet weld sizes.

1. Vertical Forces at A and B (P_{VA}, P_{VB}) Equal vertical load sharing	$P_{VA} = P_{VB} = \frac{P}{2}$$= \frac{15}{2} = 7.5\text{ kN}$$= 7500\text{ N}$
2. Horizontal Force at Top A (P_{HA}) Moment equilibrium about bottom point B	$P_{HA} \times 75 = 15 \times 50$$P_{HA} = \frac{750}{75}$$= 10\text{ kN} = 10000\text{ N}$
3. Size of Top Fillet Weld (s_1) Resultant force and top weld capacity	$P_A = \sqrt{(P_{VA})^2 + (P_{HA})^2}$$P_A = \sqrt{(7.5)^2 + (10)^2} = 12.5\text{ kN} = 12500\text{ N}$$12500 = 0.707s_1 \times 25 \times 150$$12500 = 2650s_1$$s_1 = \frac{12500}{2650}$$= 4.7\text{ mm}$
4. Size of Bottom Fillet Weld (s_2) Bottom weld designed for vertical force only	$P_{VB} = 0.707s_2 \cdot l \cdot \tau$$7500 = 0.707s_2 \times 25 \times 150$$7500 = 2650s_2$$s_2 = \frac{7500}{2650}$$= 2.83\text{ mm}$
5. Design Output Summary Shielded Arc Bracket Weld Sizes	Top Weld Size $s_1 = 4.7\text{ mm}$, Bottom Weld Size $s_2 = 2.83\text{ mm}$

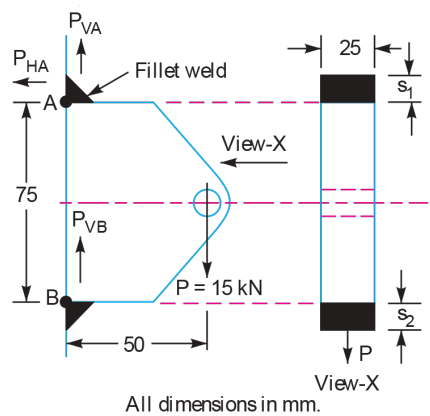


Fig. 10.34

Figure 10.34: Bracket supporting 15 kN dead weight with top and bottom fillet welds